

Alberta Data Centre Infrastructure: The “Picks & Shovels” Opportunity

A market research perspective for private equity

Aash Makwana September 2026 Lower-middle-market infrastructure services

1 Executive summary

Alberta's data centre market moved from proposal to committed capital during 2026. Meta broke ground on 8 July 2026 on a 1 GW campus in Sturgeon County representing more than C\$13 billion — its first in Canada and the province's largest confirmed digital-infrastructure commitment to date.⁷ It is paired with the C\$4.6 billion, 932 MW Greenlight Electricity Centre, a gas-fired plant developed by Pembina Pipeline, Kinetikor and Morgan Stanley Infrastructure Partners.³

Fact: the Alberta Electric System Operator (AESO) reports requested data centre load above 20,600 MW against provincial peak demand of roughly 13,000 MW,³ and its interim allowance for new large loads — 1,200 MW through 2028 — is fully allocated.^{1,12} **Interpretation:** the binding constraint is deliverable power, not customer demand. That constraint is itself the investment story: it pushes spending toward generation, electrical infrastructure and phased on-site build-out rather than real estate.

Operators themselves are a poor fit for most Canadian private equity: they are global, capital-intensive and increasingly financed through infrastructure funds and off-balance-sheet vehicles. The supply chain that builds and maintains them is not. Meta's site alone is expected to carry roughly 3,000 construction workers at peak and about 300 permanent roles.⁷ Every megawatt requires electrical distribution, switchgear, standby generation, cooling plant, controls, structured cabling and a maintenance regime running for the asset's life.

Five supporting categories look most exposed: **electrical contracting, HVAC and cooling, industrial controls and automation, connectivity and structured cabling, and critical-facility maintenance.** Alberta is unusual in that its oil-and-gas contractor base already holds most of the required trades, certifications and project-execution experience — so the relevant businesses largely exist today, serving adjacent end markets.

Investment implication: Alberta's expansion may create a secondary opportunity for specialised infrastructure and service businesses providing power, cooling, electrical, controls, connectivity, construction and maintenance services. Statistics Canada data indicates this supplier base is highly fragmented — 97.7% of Alberta building-equipment contractor establishments employ fewer than 100 people.¹³ This report does *not* conclude that any individual business is attractive, and one common assumption deserves scepticism: at the listed US comparable, recurring service and maintenance work is still only about 6% of revenue.¹⁵

2 The Alberta data centre market

Fact: in June 2025 the AESO reported 29 connection applications representing more than 16 GW of requested transmission service, and capped new large-load connections at an interim 1,200 MW through 2028 — the maximum it judged the grid could absorb without compromising reliability.¹ For scale, it noted the City of Edmonton's load is roughly 1,400 MW. The queue has since grown past 20,600 MW while that 1,200 MW remains spoken for, with reported allocations to the Greenlight-linked load and TransAlta's Keephills Phase I.^{3,12}

Announced projects and geography

Activity clusters in three corridors. In the **Alberta Industrial Heartland** north-east of Edmonton, Meta's Sturgeon County campus is under construction, with Frost Collective — a joint venture of Edmonton-headquartered Clark Builders and PCL Construction — as general contractor.⁶ In the **Calgary region**, eStruxture's CAL-3 in Rocky View County is a 205,000 sq ft, 90 MW facility supporting rack densities up to 125 kW.⁶ In the **north-west**, the O'Leary-backed Wonder Valley park in Greenview is listed by the province at \$70 billion at full build-out, but remains pre-construction.⁶ **Company announcement:** Calgary-based Beacon AI Centres has announced six Alberta campuses within a North American pipeline of 4.9 GW and forecast investment of up to C\$10 billion; these remain at planning stage.¹⁴

Policy and regulation

Government: Alberta published its AI Data Centres Strategy in December 2024, targeting up to \$100 billion of data centre and related infrastructure investment over five years.⁵ The *Data Centre Regulation* (AR 117/2026), in force 9 June 2026, codifies a “bring your own generation” (BYOG) framework under which proponents that commit offsetting generation to the common pool receive connection priority.¹⁰ The AESO published technical connection requirements on 12 June 2026, and the first BYOG intake has slipped to autumn 2026.^{2,3} A data centre levy and cost-allocation rules aim to stop these loads shifting costs onto other ratepayers.³

Alberta data centre market — key facts (September 2026)

20,600 MW	13,000 MW	1,200 MW	C\$13bn	\$100bn
Data centre load requested from the AESO AESO, via Bennett Jones, Sep 2026	Alberta system peak demand, for comparison AESO, via Bennett Jones, Sep 2026	Interim large-load connection limit to 2028 — fully allocated AESO, Jun 2025; The Hub, Sep 2026	Meta Sturgeon County campus, 1 GW, broke ground Jul 2026 Meta; Alberta Major Projects	Provincial five-year investment target (a goal, not a forecast) Government of Alberta, Dec 2024

Figures as reported by the cited sources. The \$100 billion figure is a stated government objective, not an achieved or forecast outcome.

3 Why Alberta — and what constrains it

Alberta's advantages are real but narrower than the headline narrative suggests. Each is paired below with the constraint that qualifies it.

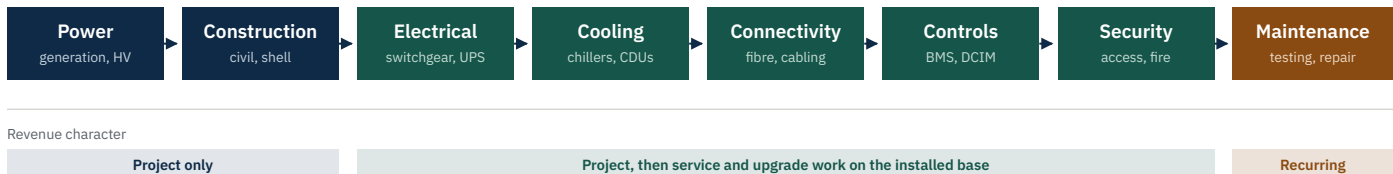
Factor	What supports development	What constrains it
Electricity market	Alberta operates Canada's only energy-only deregulated wholesale market, and the BYOG framework lets proponents self-supply rather than queue for grid capacity. ¹⁰	Grid access is the binding constraint. The interim limit is allocated, and the AESO says further large loads cannot connect without new generation, transmission or market tools. ^{1,4}
Natural gas	Low-cost gas at a discount to US benchmarks supports fast-to-build generation; Greenlight alone is expected to draw about 150 MMcf/d. ⁹	Gas pipeline and processing capacity may not scale as quickly as compute demand, creating a second bottleneck behind the grid. ¹²
Land	Large, industrially zoned parcels near existing utility corridors in Sturgeon County, Rocky View County and Greenview.	Siting is contested. Municipal zoning, not the AESO, governs land use, and proximity to residents has become decisive (Section 8). ¹¹
Climate	Cold ambient temperatures reduce cooling energy for much of the year; Meta's site uses closed-loop water cooling. ³	Rising rack densities (up to 125 kW at CAL-3) shift the problem toward liquid cooling, where the design and service base in Alberta is less established. ⁶
Industrial base	An oil-and-gas construction and maintenance workforce already holds the relevant trades — electricians, instrumentation technicians, pipefitters, millwrights — plus EPC and modular fabrication capacity. ³	That workforce is already tight. BuildForce Canada projects roughly 43,400 Alberta construction retirements by 2034, requiring more than 43,600 new hires simply to hold staffing flat. ⁸

Interpretation: the constraints do not argue against the trend; they argue about *timing and mix*. Projects that cleared the AESO's interim window are building now; everything behind them waits on BYOG, generation and transmission. The supplier opportunity is therefore likely to be lumpy and front-loaded toward power and electrical scope.

4 The data centre value chain

A data centre is assembled from eight broad scopes of work. The chain below separates project revenue (earned once, during build) from recurring revenue (earned annually thereafter) — the distinction that matters most to a buyer.

Data centre value chain — scope and revenue character



The five green stages are where an installed base converts, over time, into service revenue — the part of the chain most relevant to a buy-and-hold investor.

Stage	What the business provides and why a data centre needs it	Demand type
Power	Generation, high-voltage interconnection, substations. In Alberta this is increasingly on-site generation under BYOG rather than a grid connection.	Project
Construction	Site works, earthworks, shell and core. Largest single line item; increasingly modular and pre-fabricated to compress schedule. ³	Project
Electrical	Medium- and low-voltage distribution, switchgear, UPS, standby generators, busway, testing and commissioning. Redundancy multiplies the scope versus a conventional building.	Both
Cooling	Chillers, air handling, pumps, piping and — at AI rack densities — liquid cooling distribution units. Heat rejection is the physical limit on compute density.	Both
Connectivity	Fibre entry, dark fibre routes, structured cabling, splicing, certification testing. Carrier diversity is a design requirement, not a preference.	Both
Controls	Building management and DCIM systems, instrumentation, sensors, alarming. The layer that turns equipment into a monitored, auditable facility.	Both
Security	Access control, surveillance, fire detection and suppression. Required for customer certification and insurance.	Both
Maintenance	Preventive maintenance, load-bank and infrared testing, emergency call-out, spares. Runs for the life of the asset under contract.	Recurring

5 Where small businesses could benefit

Five categories look most directly exposed. For each, the question an investor should ask is not “does data centre demand help this business?” but “how much of the revenue it creates is durable once construction stops?”

Category	Why data centre growth could increase demand	Revenue character	Key risk
Electrical contracting	Power distribution, switchgear, standby generation and commissioning scale directly with installed MW. Redundant topologies mean far more electrical content per square foot than commercial construction.	Heavily project-weighted; service follows the installed base with a lag.	Bonding capacity and working capital. Hyperscale schedules demand scale that a \$10–30m contractor may not be able to finance.
HVAC and cooling	Chillers, pumps, piping and air handling are core scope, and AI rack densities are pushing designs toward liquid cooling that needs specialist installation and servicing.	Balanced. Mechanical plant carries genuine annual service content.	Technology shift. Firms built around air-cooled commercial HVAC may not hold the liquid-cooling capability buyers will want.
Controls and automation	Building management, instrumentation and monitoring are required for uptime guarantees and reporting. Alberta's process-automation firms already hold the relevant skill set.	Strongest recurring profile — support, licensing and upgrade cycles.	Platform dependency. Value often sits with an OEM authorisation that can be withdrawn or reassigned.
Connectivity and cabling	Fibre routes, structured cabling, splicing and certification testing at every site; carrier diversity multiplies the work per facility.	Mostly project, with modest moves-adds-changes revenue.	Low barriers in the install layer. Carriers and national integrators capture most of the economics.
Critical-facility maintenance	Preventive maintenance, testing and emergency response on power and cooling plant. Contracted, priced for response time, and renewed annually.	The genuinely recurring category.	Timing. Contracts follow commissioning, so revenue arrives years after the construction cycle begins.

Alberta companies operating in these categories

The businesses below are real Alberta companies identified through public sources, included to demonstrate that the macro trend connects to an existing local supplier base. They are **not** acquisition suggestions, and no view is offered on any of them.

Company	Location	Category	What it does and why it is relevant
PCL Construction	Edmonton	Construction	Employee-owned general contractor. Half of the Frost Collective joint venture serving as general contractor on Meta's Sturgeon County campus — the entity that will award the subcontracts smaller firms compete for. ⁶
Clark Builders	Edmonton	Construction	The other Frost Collective partner. Demonstrates that Alberta general contractors, rather than imported ones, are executing the province's largest data centre build. ⁶
Trotter & Morton	Calgary	Mechanical / electrical / facility services	Founded 1927. A multi-division group spanning mechanical and electrical construction, facility services, and electrical and controls businesses — a working Alberta example of the platform-plus-divisions structure described in Section 7.
Spartan Controls	Calgary	Controls and automation	Employee-owned since 1963; Emerson Impact Partner with 15 Western Canadian locations. Supplies process control, instrumentation and lifecycle service to power generation and industrial facilities — capabilities that transfer to data centre plant.
Finning International	Edmonton (TSX: FTT)	Standby power	The world's largest Caterpillar dealer. Supplies and services the generator sets and power systems that provide standby and, increasingly, prime power at data centre sites.
Western Electrical Management	Calgary, Edmonton	Electrical contracting	Privately held, owner-managed commercial electrical contractor operating since 1959, with a stated data and communications practice alongside new construction and service work.
EPCOR	Edmonton	Connectivity / utility	Operates a dark fibre network of 50 km or more across the Alberta Capital Region, marketed for data centre and enterprise connectivity — illustrating that fibre economics sit largely with utilities and carriers.

Interpretation: the named firms span very different scales. PCL, Finning and Spartan Controls are large enterprises; a Canadian lower-middle-market fund would not be buying them. **The lower-middle-market opportunity sits one layer below** — in the subcontractors and service firms these companies award work to, most of which are privately held, owner-operated and not publicly visible. That layer is where research would need to go next, and it is the reason a top-down screen of named Alberta companies is a poor substitute for direct sourcing.

6 Why a private equity investor would care

1. Fragmentation is measurable, not assumed

Fact: Statistics Canada data published through ISED's Canadian Industry Statistics shows 4,530 Alberta employer establishments in building equipment contracting (NAICS 2382 — electrical, mechanical and other building equipment trades) as at 2025. Of these, 2,640 employ one to four people, 1,786 employ five to 99, 102 employ 100–499, and two employ 500 or more.¹³ **Interpretation:** 97.7% of the supplier base is sub-100 employees, and only 104 establishments province-wide have the headcount to credibly staff a hyperscale scope. That is a textbook fragmented market with a very thin top end — the structure that supports platform-and-add-on strategies, though it says nothing about whether owners are willing sellers or whether price expectations are reasonable.

2. Mission-critical positioning

Data centres carry contractual uptime obligations, so equipment failure is expensive in a way that ordinary commercial buildings' is not. That favours incumbent providers with proven response times and makes switching on price alone less likely. **Interpretation:** this should support pricing power and retention — but it applies to the *service* relationship, not to competitively bid construction scope.

3. Recurring revenue — treat the claim carefully

Industry evidence: Comfort Systems USA (NYSE: FIX), the largest listed US mechanical and electrical contractor and the clearest public analogue for this thesis, reported technology work — primarily data centres — at 58% of revenue for the first half of 2026, up from 45% of full-year 2025, with backlog of \$14.06 billion at 30 June 2026 against \$8.12 billion a year earlier. In the same disclosure, service and maintenance accounted for roughly 6% of revenue and service projects a further 5%.¹⁵ **Interpretation:** the data centre boom is, so far, overwhelmingly a project-revenue phenomenon. Recurring maintenance income builds slowly behind the installed base. An investor underwriting “recurring revenue” in this sector should verify the actual mix rather than assume it.

4. Technical and regulatory barriers to entry

Barriers are real but sit in specific places: Alberta Safety Codes Council trade certification and municipal contractor licensing; Red Seal journeyman supply; bonding and insurance capacity; and OEM authorisations such as a Caterpillar dealership or an Emerson Impact Partner designation, which are granted rather than bought. **Interpretation:** the durable moat is usually the combination of certified crews, a safety record acceptable to hyperscale clients, and an OEM relationship — not any single one of them.

5. Cross-selling from an existing industrial base

Alberta contractors serving oil and gas already perform medium-voltage electrical work, instrumentation, pipe fitting and turnaround maintenance under comparable safety regimes. **Interpretation:** the transferability is genuine, lowering execution risk — but it also means the capability is not scarce locally, which caps what any single firm can charge for it.

6. Consolidation potential

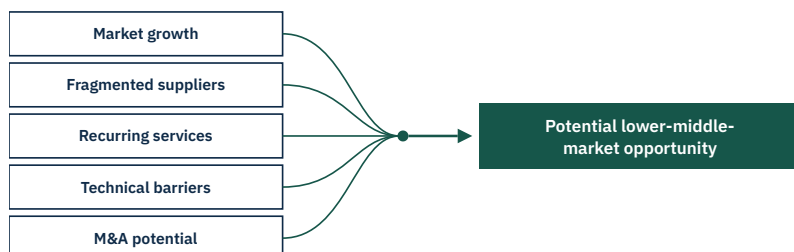
The preconditions for consolidation are present: a fragmented base, an ageing ownership cohort, and buyers who value regional density. Comfort Systems' record shows the model can work at scale in the US.¹⁵ **Interpretation:** this does not mean consolidation will occur in Alberta. Owner-operators in a strong market have little pressure to sell, trade labour constrains integration benefits, and a roll-up assembled at the top of a capex cycle carries obvious timing risk.

7 Illustrative opportunity structure

Hypothetical — illustrative only

A sponsor acquires a regional Alberta industrial electrical contractor that has historically served oil-and-gas and commercial customers. The company wins data-centre-related subcontracts, then acquires two smaller specialist contractors in adjacent Alberta markets to add controls and commissioning capability and to extend geographic coverage. No purchase price, multiple, capital structure or return is assumed or implied, and no such transaction is known to the author.

Why the structure could be attractive — and what has to hold



Each input depends on an assumption that must be tested

Announced megawatts convert into built megawatts
 Owners are willing sellers at defensible prices
 Service mix is material, not a rounding error
 Certified crews can actually be retained
 Integration benefits survive a tight labour market

Framework only. Not a recommendation, and not evidence that any of these conditions currently hold in Alberta.

8 Key risks

- 1. Projects are being delayed and refused, not just built.** **Fact:** on 17 August 2026 the Alberta Utilities Commission denied Synapse Real Estate Corp.'s application for a 1,400 MW gas plant intended to power a roughly \$10 billion data centre campus at Olds, finding the site “too close to the surrounding community” and that the company had not justified the location. Around 685 parties had sought standing.¹¹ Announced megawatts are not contracted megawatts.
- 2. Grid capacity limits the near-term pipeline.** The AESO's 1,200 MW interim allowance is allocated, the first BYOG intake has slipped to autumn 2026, and further connections depend on new generation or transmission.^{1,3} Supplier revenue could be concentrated in a small number of projects for several years.
- 3. A second bottleneck in natural gas.** Gas pipeline and supply capacity may not expand as fast as generation demand, which would push back the in-service dates that supplier backlogs are built on.¹² Greenlight is not expected in service until late 2030.⁹
- 4. Skilled labour is the operating constraint.** BuildForce Canada projects roughly 43,400 Alberta construction retirements by 2034 against a need for more than 43,600 new hires.⁸ Wage inflation compresses fixed-price margins, and in a labour-constrained market acquiring a competitor may buy the backlog without buying the capacity to deliver it.
- 5. Customer concentration.** A contractor that wins meaningful hyperscale scope can find a single client at 40–60% of revenue within two years. That is a valuation discount, not a growth story, and it inverts the bargaining power that mission-critical positioning is supposed to confer.
- 6. The capital-expenditure cycle.** AI infrastructure spending is set by a handful of global buyers and has moved faster than any prior data centre cycle. Comfort Systems' backlog rose from \$8.12 billion to \$14.06 billion in twelve months¹⁵ — a rate of change that cuts both ways. Assets bought on peak-cycle multiples with peak-cycle margins carry obvious risk.
- 7. Alberta-specific policy uncertainty.** Cost-allocation rules, the data centre levy and BYOG mechanics are still settling,^{3,10} and a senior AWS executive has publicly argued that the absence of a federal text-and-data-mining exception in Canadian copyright law effectively excludes AI *training* workloads from Canada — potentially limiting Alberta to lower-intensity inference and storage.³

9 What an investor would need to verify

- 1 What share of revenue is genuinely exposed to data centres today, versus scope that is merely adjacent or aspirational?
- 2 What is the split between one-time project revenue and contracted recurring service revenue, measured from contracts rather than management estimates?
- 3 What percentage of revenue comes from the top five customers, and how does that change if the largest current project completes on schedule?
- 4 What is the backlog, how much of it is signed versus verbally awarded, and what are the cancellation and re-pricing provisions?
- 5 What are EBITDA margins by contract type, and how have fixed-price jobs performed against bid over the last three years?
- 6 How dependent is delivery on specific certified individuals, what is journeyman turnover, and what is the apprentice pipeline?
- 7 What OEM authorisations, safety qualifications and client pre-qualifications does the business hold, and can they survive a change of control?
- 8 Does management have credible acquisition and integration experience, and can the business win work outside Alberta if the provincial cycle turns?

10 Conclusion

Alberta's data centre development is now underway rather than prospective, and it creates potential secondary demand across a range of infrastructure and service providers. The scale is meaningful: a single confirmed project carries more than C\$13 billion of capital expenditure and about 3,000 peak construction workers,⁷ and requested load in the AESO queue exceeds the province's entire peak demand.³ Equally, the constraints are real, and the Olds refusal is a reminder that a proposal listed in a provincial database is not a contract.

From a research perspective, the more interesting businesses are not the data centre operators — which are globally financed and largely out of reach for Canadian lower-middle-market capital — but the specialised electrical, mechanical, controls, connectivity and maintenance companies that supply them. Alberta's supplier base is measurably fragmented,¹³ and its industrial heritage means the required capabilities already exist locally rather than needing to be imported.

The question that would determine whether any of this is investable is narrower than the macro trend suggests: do these businesses actually possess recurring revenue, technical barriers, customer stickiness and a credible path to organic and acquisition-driven growth — or does the data centre cycle simply deliver a large volume of competitively bid, fixed-price project work into a labour-constrained market? The public evidence available today is not sufficient to answer that question, and answering it would require company-level diligence of the kind set out in Section 9. That is where further research should go.

— Sources

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Company descriptions in Section 5 draw on the companies’ own public websites and the Alberta Major Projects inventory. All financial figures are in Canadian dollars unless marked otherwise. Statements are labelled throughout as fact, company announcement, industry evidence or interpretation; government targets are identified as objectives rather than forecasts. This document is market research prepared as an analytical exercise. It is not investment advice, does not constitute an offer or recommendation in respect of any security or company, and no company named should be inferred to be an acquisition target.